

MAT 110

Lecture 04

Chain Rule, Technique of Differentiation

Akram Hosen (RMH)

Lecturer, MPS.

BRAC University

Contents

1	Chain Rule of Differential Calculus	1
2	Explicit and Implicit Functions	3
3	Implicit Differentiation	4
4	Logarithmic Differentiation	5
5	Parametric Differentiation	7
6	Differentiation with Respect to Another Function	8
7	Using Trigonometric Identities Before Differentiation	9
8	Practice Problems: Techniques of Differentiation	10

1 Chain Rule of Differential Calculus

Chain Rule

If

$$y = f(u), \quad u = g(x),$$

then

$$\frac{dy}{du} = f'(u), \quad \frac{du}{dx} = g'(x),$$

and

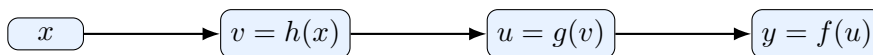
$$\boxed{\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}}.$$

More generally, if

$$y = f(u), \quad u = g(v), \quad v = h(x),$$

then

$$\boxed{\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}}.$$



Chain of dependency: $x \rightarrow v \rightarrow u \rightarrow y$.

Example 1: Derivative using chain rule

Evaluate

$$\frac{dy}{dx} \quad \text{where} \quad y = \ln(\cos(x^3)).$$

Answer.

Given

$$y = \ln(\cos(x^3)).$$

Let

$$y = \ln u, \quad u = \cos v, \quad v = x^3.$$

Then

$$\frac{dy}{du} = \frac{1}{u}, \quad \frac{du}{dv} = -\sin v, \quad \frac{dv}{dx} = 3x^2.$$

Using the chain rule,

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx} = \frac{1}{u}(-\sin v)(3x^2).$$

Now substitute

$$u = \cos(x^3), \quad v = x^3.$$

Hence

$$\frac{dy}{dx} = \frac{1}{\cos(x^3)} \{-\sin(x^3)\} 3x^2 = -3x^2 \frac{\sin(x^3)}{\cos(x^3)}.$$

Therefore

$$\boxed{\frac{dy}{dx} = -3x^2 \tan(x^3)}.$$

Example 2: Derivative of $\tan^2 x$

Evaluate

$$\frac{dy}{dx} \quad \text{where} \quad y = \tan^2 x.$$

Answer.

Given

$$y = \tan^2 x = (\tan x)^2.$$

Let

$$u = \tan x.$$

Then

$$y = u^2, \quad u = \tan x.$$

Now

$$\frac{dy}{du} = 2u, \quad \frac{du}{dx} = \sec^2 x.$$

Therefore

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = 2u \sec^2 x.$$

Putting $u = \tan x$, we get

$$\boxed{\frac{dy}{dx} = 2 \tan x \sec^2 x}.$$

In short form,

$$\frac{d}{dx}(\tan x)^2 = 2(\tan x) \cdot \frac{d}{dx}(\tan x) = 2 \tan x \sec^2 x.$$

Example 3: Derivative of logarithmic functions

Evaluate the following derivatives:

$$\frac{d}{dx}(\ln x), \quad \frac{d}{dx}[\ln(x^2 + 1)].$$

Answer.

First,

$$\boxed{\frac{d}{dx}(\ln x) = \frac{1}{x}}, \quad x > 0.$$

For the second derivative, let

$$u = x^2 + 1.$$

Then

$$\frac{d}{dx} \ln(x^2 + 1) = \frac{d}{du}(\ln u) \cdot \frac{du}{dx} = \frac{1}{u} \cdot 2x.$$

Substituting $u = x^2 + 1$, we get

$$\boxed{\frac{d}{dx} \ln(x^2 + 1) = \frac{2x}{x^2 + 1}}.$$

Example 4: Composite function with power and trigonometric functions

If

$$y = (1 + \cos(x^2 - 1))^{3/2},$$

evaluate $\frac{dy}{dx}$.

Answer.

Given

$$y = (1 + \cos(x^2 - 1))^{3/2}.$$

We differentiate step by step:

$$\frac{dy}{dx} = \frac{d}{dx} (1 + \cos(x^2 - 1))^{3/2}.$$

By the chain rule,

$$\frac{dy}{dx} = \frac{3}{2} (1 + \cos(x^2 - 1))^{1/2} \cdot \frac{d}{dx} (1 + \cos(x^2 - 1)).$$

Now

$$\frac{d}{dx} (1 + \cos(x^2 - 1)) = 0 - \sin(x^2 - 1) \cdot \frac{d}{dx} (x^2 - 1).$$

Since

$$\frac{d}{dx} (x^2 - 1) = 2x,$$

we get

$$\frac{d}{dx} (1 + \cos(x^2 - 1)) = -2x \sin(x^2 - 1).$$

Therefore

$$\frac{dy}{dx} = \frac{3}{2} (1 + \cos(x^2 - 1))^{1/2} (-2x \sin(x^2 - 1)).$$

Hence

$$\boxed{\frac{dy}{dx} = -3x \sin(x^2 - 1) (1 + \cos(x^2 - 1))^{1/2}}.$$

2 Explicit and Implicit Functions

Explicit function

An equation of the form

$$y = f(x)$$

is said to define y explicitly as a function of x , because y appears alone on one side of the equation and does not appear on the other side.

Examples:

$$y = \frac{x-1}{x+1}, \quad y = \sin x.$$

Implicit function

An **implicit function** is a function in which x and y are related by an equation, but y is not isolated as a single explicit function of x .

Example:

$$5y^2 + \sin y = x^2.$$

Here both x and y occur in the same equation, and y is not written alone in the form $y = f(x)$.

3 Implicit Differentiation

Idea of implicit differentiation

When y is a function of x , every occurrence of y must be treated as $y(x)$. Therefore, when differentiating terms involving y , we must use the chain rule. For example,

$$\frac{d}{dx}(y^2) = 2y \frac{dy}{dx}, \quad \frac{d}{dx}(\sin y) = \cos y \frac{dy}{dx}.$$

Example 5: Implicit differentiation

Find $\frac{dy}{dx}$ if

$$5y^2 + \sin y = x^2.$$

Answer.

Given

$$5y^2 + \sin y = x^2.$$

Differentiate both sides with respect to x :

$$\frac{d}{dx}(5y^2 + \sin y) = \frac{d}{dx}(x^2).$$

Then

$$5 \frac{d}{dx}(y^2) + \frac{d}{dx}(\sin y) = 2x.$$

Using the chain rule,

$$5 \cdot 2y \frac{dy}{dx} + \cos y \frac{dy}{dx} = 2x.$$

So

$$10y \frac{dy}{dx} + \cos y \frac{dy}{dx} = 2x.$$

Factor out $\frac{dy}{dx}$:

$$\frac{dy}{dx}(10y + \cos y) = 2x.$$

Therefore

$$\boxed{\frac{dy}{dx} = \frac{2x}{10y + \cos y}}.$$

4 Logarithmic Differentiation

Logarithmic differentiation

Logarithmic differentiation is a useful technique for differentiating functions where the variable appears both in the base and in the exponent.

If

$$y = (f(x))^{g(x)},$$

then we first take logarithm on both sides:

$$\ln y = g(x) \ln(f(x)).$$

Then we differentiate both sides with respect to x and finally multiply by y .

Important formula

If

$$y = (f(x))^{g(x)},$$

then

$$\ln y = g(x) \ln(f(x)).$$

Differentiating,

$$\frac{1}{y} \frac{dy}{dx} = g'(x) \ln(f(x)) + g(x) \frac{f'(x)}{f(x)}.$$

Therefore,

$$\frac{dy}{dx} = (f(x))^{g(x)} \left[g'(x) \ln(f(x)) + g(x) \frac{f'(x)}{f(x)} \right].$$

Example 1: Differential coefficient of $(\sin x)^{\ln x}$

Find the differential coefficient of

$$y = (\sin x)^{\ln x}.$$

Answer.

Given

$$y = (\sin x)^{\ln x}.$$

Taking logarithm on both sides,

$$\ln y = \ln x \cdot \ln(\sin x).$$

Differentiating both sides with respect to x ,

$$\frac{d}{dx}(\ln y) = \frac{d}{dx} \{ \ln x \cdot \ln(\sin x) \}.$$

So,

$$\frac{1}{y} \frac{dy}{dx} = \ln x \cdot \frac{1}{\sin x} \cos x + \ln(\sin x) \cdot \frac{1}{x}.$$

Hence,

$$\frac{1}{y} \frac{dy}{dx} = \ln x \cot x + \frac{1}{x} \ln(\sin x).$$

Therefore,

$$\frac{dy}{dx} = y \left\{ \ln x \cot x + \frac{1}{x} \ln(\sin x) \right\}.$$

Since

$$y = (\sin x)^{\ln x},$$

we get

$$\boxed{\frac{d}{dx}(\sin x)^{\ln x} = (\sin x)^{\ln x} \left\{ \ln x \cot x + \frac{1}{x} \ln(\sin x) \right\}}.$$

Example 2: Sum of two variable-base variable-exponent functions

Let

$$y = (\sin x)^{\cos x} + (\cos x)^{\sin x}.$$

Find

$$\frac{dy}{dx}.$$

Answer.

Let

$$y = u + v,$$

where

$$u = (\sin x)^{\cos x} \quad \text{and} \quad v = (\cos x)^{\sin x}.$$

Then

$$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}. \quad (1)$$

First,

$$u = (\sin x)^{\cos x}.$$

Taking logarithm,

$$\ln u = \cos x \ln(\sin x).$$

Differentiating both sides,

$$\frac{1}{u} \frac{du}{dx} = \cos x \cdot \frac{\cos x}{\sin x} + \ln(\sin x)(-\sin x).$$

So,

$$\frac{1}{u} \frac{du}{dx} = \cos x \cot x - \sin x \ln(\sin x).$$

Therefore,

$$\frac{du}{dx} = (\sin x)^{\cos x} \{ \cos x \cot x - \sin x \ln(\sin x) \}. \quad (2)$$

Again,

$$v = (\cos x)^{\sin x}.$$

Taking logarithm,

$$\ln v = \sin x \ln(\cos x).$$

Differentiating both sides,

$$\frac{1}{v} \frac{dv}{dx} = \sin x \cdot \frac{-\sin x}{\cos x} + \ln(\cos x) \cos x.$$

So,

$$\frac{1}{v} \frac{dv}{dx} = -\sin x \tan x + \cos x \ln(\cos x).$$

Therefore,

$$\frac{dv}{dx} = (\cos x)^{\sin x} \{ \cos x \ln(\cos x) - \sin x \tan x \}. \quad (3)$$

From (1), (2), and (3),

$$\boxed{\frac{dy}{dx} = (\sin x)^{\cos x} \{ \cos x \cot x - \sin x \ln(\sin x) \} + (\cos x)^{\sin x} \{ \cos x \ln(\cos x) - \sin x \tan x \}}.$$

5 Parametric Differentiation

Parametric equation

If x and y are both expressed in terms of a third variable, say θ or t , then the equations are called parametric equations.

For example,

$$x = f(\theta), \quad y = g(\theta).$$

Here θ is called a parameter.

Derivative for parametric equations

If

$$x = f(\theta), \quad y = g(\theta),$$

then

$$\boxed{\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}}$$

provided that

$$\frac{dx}{d\theta} \neq 0.$$

Example 3: Parametric differentiation

Find

$$\frac{dy}{dx}$$

where

$$x = a \cos^3 \theta, \quad y = a \sin^3 \theta.$$

Answer.

Given

$$x = a \cos^3 \theta, \quad y = a \sin^3 \theta.$$

Now,

$$\frac{dx}{d\theta} = a \cdot 3 \cos^2 \theta (-\sin \theta) = -3a \cos^2 \theta \sin \theta.$$

Also,

$$\frac{dy}{d\theta} = a \cdot 3 \sin^2 \theta \cos \theta = 3a \sin^2 \theta \cos \theta.$$

Therefore,

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{3a \sin^2 \theta \cos \theta}{-3a \cos^2 \theta \sin \theta}.$$

So,

$$\frac{dy}{dx} = -\frac{\sin \theta}{\cos \theta}.$$

Hence,

$$\boxed{\frac{dy}{dx} = -\tan \theta}.$$

6 Differentiation with Respect to Another Function

Differentiation with respect to another function

If y and z are both functions of x , then the derivative of y with respect to z is

$$\boxed{\frac{dy}{dz} = \frac{\frac{dy}{dx}}{\frac{dz}{dx}}}$$

provided that

$$\frac{dz}{dx} \neq 0.$$

Example 4: Differentiation with respect to another function

Differentiate

$$\cos^{-1} \left(\frac{1-x^2}{1+x^2} \right)$$

with respect to

$$\tan^{-1} \left(\frac{2x}{1-x^2} \right).$$

Answer.

Let

$$y = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right)$$

and

$$z = \tan^{-1} \left(\frac{2x}{1-x^2} \right).$$

Using the standard identities,

$$\cos(2 \tan^{-1} x) = \frac{1-x^2}{1+x^2}$$

and

$$\tan(2 \tan^{-1} x) = \frac{2x}{1-x^2},$$

we get

$$\cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) = 2 \tan^{-1} x$$

and

$$\tan^{-1} \left(\frac{2x}{1-x^2} \right) = 2 \tan^{-1} x.$$

Thus,

$$y = 2 \tan^{-1} x, \quad z = 2 \tan^{-1} x.$$

Therefore,

$$\frac{dy}{dz} = \frac{\frac{dy}{dx}}{\frac{dz}{dx}} = \frac{\frac{d}{dx}(2 \tan^{-1} x)}{\frac{d}{dx}(2 \tan^{-1} x)} = 1.$$

Hence,

$$\boxed{\frac{d}{dz} \left[\cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) \right] = 1}.$$

7 Using Trigonometric Identities Before Differentiation

Important idea

Sometimes a given inverse trigonometric expression can be simplified before differentiating. For example,

$$\tan^{-1} \left(\frac{1 - \tan x}{1 + \tan x} \right) = \tan^{-1} \{ \tan(\frac{\pi}{4} - x) \}.$$

So, under the usual principal-value assumption,

$$\tan^{-1} \{ \tan(\frac{\pi}{4} - x) \} = \frac{\pi}{4} - x.$$

Example 5: Simplification before differentiation

If

$$y = \tan^{-1} \left(\frac{\cos x - \sin x}{\cos x + \sin x} \right),$$

then find

$$\frac{dy}{dx}.$$

Answer.

Given

$$y = \tan^{-1} \left(\frac{\cos x - \sin x}{\cos x + \sin x} \right).$$

Divide numerator and denominator by $\cos x$:

$$y = \tan^{-1} \left(\frac{1 - \tan x}{1 + \tan x} \right).$$

Now,

$$\tan \left(\frac{\pi}{4} - x \right) = \frac{1 - \tan x}{1 + \tan x}.$$

Therefore,

$$y = \tan^{-1} \left[\tan \left(\frac{\pi}{4} - x \right) \right].$$

Hence,

$$y = \frac{\pi}{4} - x.$$

Differentiating with respect to x ,

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{\pi}{4} - x \right) = -1.$$

Therefore,

$$\boxed{\frac{dy}{dx} = -1}.$$

8 Practice Problems: Techniques of Differentiation

1. Find the differential coefficients of the following functions with respect to x (i.e. $\frac{dy}{dx}$).

$$\begin{aligned} (i) \ y &= \sin x \sin 2x \sin 3x, & (ii) \ y &= \sec^3 x, & (iii) \ y &= \cos 2x \cos 3x, \\ (iv) \ y &= \sin^{-1}(x^2), & (v) \ y &= \tan(\sin^{-1} x), & (vi) \ y &= \cot^{-1} \left(\frac{1+x}{1-x} \right), \\ (vii) \ y &= \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right), & (viii) \ y &= \sin^{-1} \left(\frac{2x}{1+x^2} \right), & (ix) \ y &= \tan^{-1} \left(\frac{2x}{1-x^2} \right), \\ (x) \ y &= \tan^{-1} \left(\frac{x}{\sqrt{1-x^2}} \right). \end{aligned}$$

2. Find the differential coefficients of:

$$(i) \ (\sin x)^{\ln x}, \quad (ii) \ (\sin x)^{\cos x} + (\cos x)^{\sin x}.$$

3. Find the differential coefficient of $\sqrt{\frac{1+x}{1-x}}$.

4. Find $\frac{dy}{dx}$ in the following cases:

$$\begin{aligned} (i) \ 3x^4 - x^2y + 2y^3 &= 0, & (ii) \ x^3 + y^3 + 4x^2y - 25 &= 0, \\ (iii) \ x^y &= y^x. \end{aligned}$$

5. Find $\frac{dy}{dx}$ when

$$\begin{aligned} (i) \ x &= a \cos^3 \theta, \quad y = a \sin^3 \theta, \\ (ii) \ x &= \sin^2 \theta, \quad y = \tan \theta, \\ (iii) \ x &= a \sec^2 \theta, \quad y = a \tan^2 \theta. \end{aligned}$$

6. Differentiate the left-side functions with respect to the right-side ones:

$$(i) \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) \quad \text{with respect to} \quad \tan^{-1} \left(\frac{2x}{1-x^2} \right),$$

$$(ii) x^{\sin^{-1} x} \quad \text{with respect to} \quad \sin^{-1} x.$$

7. Find the differential coefficients of:

$$(i) \ln \sqrt{\frac{1+\sin x}{1-\sin x}}, \quad (ii) \ln \sqrt{\frac{1-\cos x}{1+\cos x}},$$

$$(iii) \tan^{-1} \sqrt{\frac{1-\cos x}{1+\cos x}}, \quad (iv) \tan^{-1} \left(\frac{\cos x - \sin x}{\cos x + \sin x} \right),$$

$$(v) \sin \left(2 \tan^{-1} \sqrt{\frac{1-x}{1+x}} \right).$$